Lecture-28
External Flows I: Fundamental Concepts

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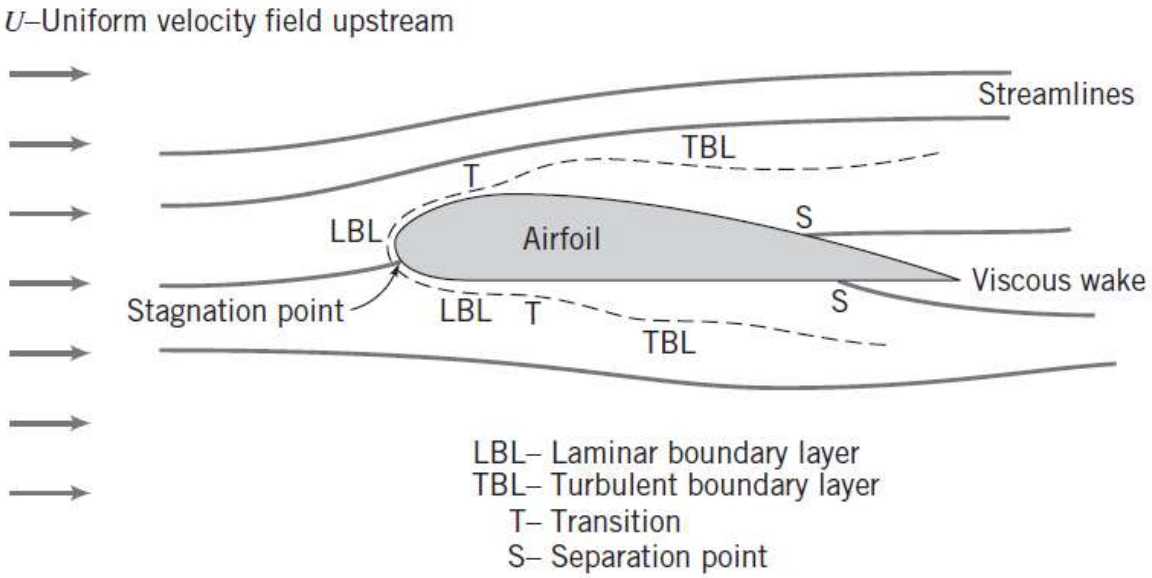
Learning Objectives

By the end of this lecture, you should be able to

1. Locate and define **external flow features** on an immersed body, including the **stagnation point**, **transition point**, and **viscous wake**.
2. Contrast the **structure**, **profiles**, and **wall shear stress** behavior between laminar and turbulent boundary layers.
3. Calculate and physically interpret **boundary layer thickness**, **displacement thickness**, and **momentum thickness**.

Basic ideas about external flows

- External flows are **flows over bodies immersed** in an unbounded fluid.

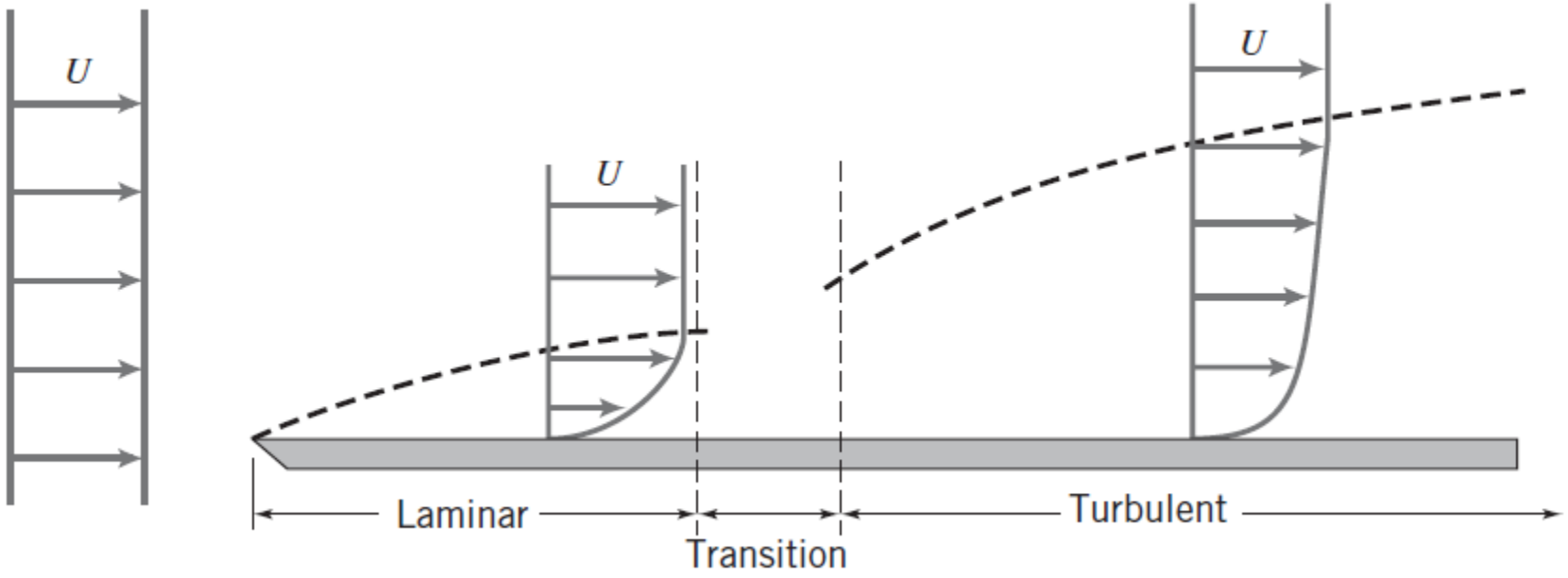


- Transition to turbulent flow **occurs at some distance** from the stagnation point.

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Boundary Layers

- Many viscous flows can be analyzed by **dividing the flow** into a region **close to the solid boundaries** and another covering the **rest of the flow**.
- The flow inside the boundary layer might be **laminar or turbulent**.
- For calculations, the **transition Reynolds number** is usually taken as **$5X10^5$**.



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Boundary Layers Thickness

- Locating the **edge of the boundary layer** is not straightforward.
- **Boundary layer thickness**, **displacement thickness,** and **momentum thickness** are three concepts used to quantify the boundary layer thickness.
- **BL thickness** is the distance at which the velocity becomes **99% of the free-stream velocity**.
- **Displacement thickness** is the distance the plate should be moved so that the **loss of mass flux** due to the reduction in velocity **is compensated**.
- **Momentum thickness** is the distance the plate should be moved so that the **loss of momentum flux** due to the reduction in velocity **is compensated**.

$$\delta^* = \int_0^\infty \left(1-\frac{u}{U}\right)dy \approx \int_0^\delta \left(1-\frac{u}{U}\right)dy$$

$$\theta = \int_0^\infty \frac{u}{U}\left(1-\frac{u}{U}\right)dy \approx \int_0^\delta \frac{u}{U}\left(1-\frac{u}{U}\right)dy$$

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Revisiting the learning objectives

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